Questions

Q1.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

This question is about radioactivity.

Alpha (α), beta (β) and gamma (γ) are three types of radioactive emissions.

Which statement describes **all** of these radioactive emissions?

(1)

☐ **A** ionising and emitted by stable nuclei

☐ **B** ionising and emitted by unstable nuclei

☐ **C** neutral and emitted by stable nuclei

☐ **D** neutral and emitted by unstable nuclei

(Total for question = 1 mark)

Q2.

Nuclear fusion provides the energy source for stars including the Sun.

Describe what happens during nuclear fusion.

(3)

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(Total for question = 3 marks)

Q3.

Describe what happens in the nucleus of an atom when a beta plus particle is emitted.

(2)

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(Total for question = 2 marks)

Q4.

Describe what happens in the nucleus of an atom when a positron is emitted.

(2)

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(Total for question = 2 marks)

Q5.

In a nuclear fusion reaction, mass is converted into energy.

(i) Describe what is meant by nuclear fusion.

(2)

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(ii) The speed of light, c, is 3.0×10^8 m/s.

The energy radiated by the Sun in 1 second is 3.9×10^{26} J.
Calculate the decrease in the Sun's mass equivalent to this energy.
State the unit with your answer.
Use the equation

$$E = m \times c^2$$

(2)

decrease in mass = ...

(Total for question = 4 marks)

Q6.

Carbon–14 is radioactive and has a half-life of 5 700 years.

The number of radioactive carbon–14 atoms in a very old piece of wood is found to have decreased from 1 000 000 to 125 000.

Determine the age of the piece of wood.

(2)

age of wood = ... years

(Total for question = 2 marks)

Q7.

This question is about radioactivity.

Fluorine-19 is a stable isotope of the element fluorine.

The element fluorine also has several radioactive isotopes.

Describe **one** similarity and **one** difference between the numbers of particles in one nucleus of fluorine-19 and one nucleus of a radioactive isotope of fluorine.

(2)

similarity

.....

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difference

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(Total for question = 2 marks)

Q8.

*The global demand for electricity is increasing.

There is a debate about whether nuclear power generation should or should not contribute to meeting this increasing demand.

Discuss the arguments for and against using nuclear power to meet the increasing global demand for electricity.

(6)

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(Total for question = 6 marks)

Q9.

Figure 4 shows a Geiger-Müller (GM) tube used for measuring radioactivity.

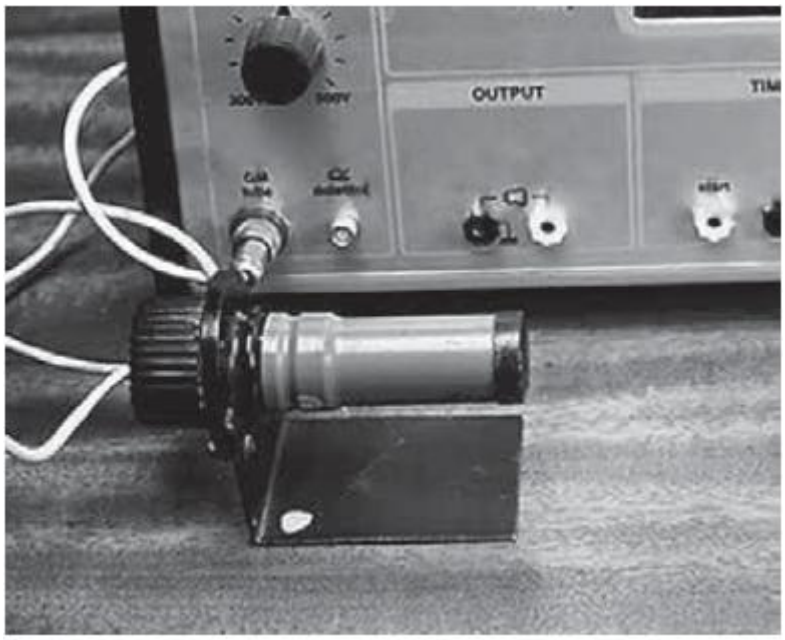

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Figure 4

One radioactive source used in hospitals is technetium (Tc).

Technetium is produced from the radioactive decay of molybdenum (Mo).

Complete the following nuclear equation.

(1)

$$^{99}_{42}\text{Mo} \rightarrow \underline{\quad}\,\text{Tc} + {}^{0}_{-1}\beta$$

(Total for question = 1 mark)

Q10.

Rutherford devised an experiment to fire alpha particles at thin gold foil.

It was found that alpha particles were scattered by the gold foil.

The gold foil was about 4.0×10^{-7}m thick.

A gold atom has a diameter of about 0.15 nm.

Estimate how many gold atoms would fit across this thickness of gold foil.

(2)

number of atoms = ...

(Total for question = 2 marks)

Q11.

The half-life of cobalt-60 is 5 years.

A school cobalt source had an activity of 38.5 kBq in the year 2000.

Estimate the activity of this source in the year 2020.

(3)

activity = ... kBq

(Total for question = 3 marks)

Mark Scheme

Q1.

Question number	Answer	Mark
CS2	B ionising and emitted by unstable nuclei A is incorrect stable nuclei do not give radioactive emissions C is incorrect not all radioactive emissions are neutral D is incorrect not all radioactive emissions are neutral	(1) AO1

Q2.

Question number	Answer	Additional guidance	Mark
	A description to include any three of the following		(3)
	(smaller) nuclei / atoms / particles (1)	two named eg hydrogen (nuclei)	
	come together / join (1)	allow fuse not 'bond'	
	to produce a larger nucleus / atom / particle (1)	helium for nucleus	
	needing high temperature / pressure (1)	accept fast (moving) nuclei	
	overcoming repulsion (between nuclei) (1)		
	energy released (1)	ignore energy created	

Q3.

Question Number	Answer	Additional guidance	Mark
	description to include proton (1) becomes a neutron (1)	 charge decreases by 1 atomic number decreases by 1 mass (number) stays the same neutron number increases by 1 award 2 marks for p → n award both marks for answers in terms of quarks: up → down or u → d or uud → udd if no other mark scored award 1 mark for n → p	**(2)** **AO1.1**

Q4.

	Answer	Additional guidance	Mark
	description to include: proton (1) becomes a neutron (1)	 award 2 marks for p → n award 2 marks for answers in terms of quarks: u→d or up → down or uud→udd if no other mark scored, allow 1 mark for any **one** of neutron becomes proton n → p d→u decrease in atomic number <u>by one</u> mass number stays the same gains a neutron reduce charge (of nucleus) <u>by one</u> responses referring to emission of gamma or neutrino	**(2)** **AO1**

Q5.

Question Number	Answer	Additional guidance	Mark
(i)	description to include (small) nuclei join / fuse / combine (1) to create larger nuclei (1)	do not accept atoms / particles for nuclei accept named nuclei hydrogen nuclei join to become helium (nuclei)	**(2)** **AO1.1**

Question Number	Answer	Additional guidance	Mark
(ii)	substitution and rearrangement (1) $(m =)\ \frac{3.9 \times 10^{26}}{[3.0 \times 10^{8}]^{2}}$ evaluation and unit(1) 4.3×10^{9} kg	$(m =)\ \frac{3.9 \times 10^{26}}{9.0 \times 10^{16}}$ allow values that round to 4.3×10^{9} kg e.g. 4.333×10^{9} kg 4.3×10^{9} with no unit or the wrong unit scores 1 mark award full marks for the correct answer without working	**(2)** **AO2.1**

Q6.

Question Number	Answer	Additional guidance	Mark
	processing (1) $\frac{125\ 000}{1\ 000\ 000}$ OR $\frac{1}{8}$ OR 3 half-lives or 3 x 5700 evaluation (1) 17 100	 accept an appropriate attempt using more than one halving 17 000 award full marks for the correct answer without working	**(2)**

Q7.

Question number	Answer	Additional guidance	Mark
CS2	same number of protons (1) different number of neutrons (1)	same atomic number different mass number	**(2)** **AO2**

Q8.

Question number	Indicative content	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive, and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. AO1 1 **(6 marks)** <u>Arguments for using nuclear power</u> • we are running out of fossil fuels SO the use of nuclear can play its part where we have previously used fossil fuels • no carbon emissions SO does not contribute towards global warming, rise in sea-levels etc. • low running costs SO a good economical alternative to other power sources • with reprocessing can last hundreds of years • large power output compared with wind farms / solar etc. <u>Arguments against using nuclear power</u> • expensive to set up and decommission SO the use of other sources could be more economical • nuclear disasters possible SO there could be a loss of life / contamination of an area • nuclear toxic waste lasts a long time • waste storage issues – contamination of ground water • terrorism / bomb making	**(6)Exp**

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

SUMMARY, for guidance			
Level	**Mark**	**Additional Guidance**	**General additional guidance – the decision within levels** e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance Unbalanced incomplete argument Elements of physics present isolated knowledge for or against	Possible candidate responses no CO_2 emissions high cost
Level 2	3–4	Additional guidance Argument linked with evidence, but not entirely coherent or balanced Uses evidence for **or** against but lacks balance / focus / with logical connections made on one side and statement from the other	Possible candidate responses no CO_2 emissions so no contribution to global warming large power output compared with wind farms
Level 3	5–6	Additional guidance Developed detailed arguments, including use of evidence, advanced both for **and** against the use of nuclear power	Possible candidate responses no CO_2 emissions so no contribution to global warming large power output compared with wind farms nuclear toxic waste lasts a long time

Q9.

Question Number	Answer	Mark
	an answer containing both of the following numbers in the correct places (1) $^{99}_{43}Tc$	**(1)** AO 2 1

Q10.

	Answer	Additional guidance	Mark
	substitution (1) number of atoms = $\frac{4.0\times10^{-7}}{0.15\ (\times10^{-9})}$ evaluation (1) 2 700	$\frac{4.0\times10^{-7}}{1.5\ (\times10^{-10})}$ $\frac{0.000\ 000\ 4}{0.000\ 000\ 000\ 15}$ accept any value that rounds to 2 700 award full marks for correct answer without working	**(2)** **AO2**

Q11.

Question number	Answer	Additional guidance	Mark
	recognition of there being **4 half lives** involved (1) so fraction of **1/16** involved (1) evaluation (1) 2.4 (kBq)	allow 2 marks for 4.8 (kBq) (used three instead of 4 half lives) allow 1 mark for any other $(1/2)^n$ being involved i.e. for answers that round to 19.3 (kBq), 9.63 (kBq), 1.2(kBq) award full marks for the correct answer without working	**(3)**